

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C	A,B,C,D	A,B,C	C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	A	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.29 to 1.31	2.50	0.50	9.00	1.84 to 1.86	9.00	6.00	4.68

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,D	B,D	A,B,C,D	A,B,C,D	B,C	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	A	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	754.00	391.30	8.00	552.00	548.00	8.00	23.80	4.32

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C,D	A,B	A,B,C	A,C	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	C	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	177.00	100.00	667.00	200.00	6.00	251.00	5.00	0.33 or 0.34

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A,B,C,D)

$$F' = \left(\frac{1}{\alpha\beta}\right) F \Rightarrow m'a' = \left(\frac{1}{\alpha\beta}\right) ma$$

$$\Rightarrow m' = \left(\frac{1}{\alpha\beta}\right) \left(\frac{a}{a'}\right) m$$

$$\text{But } \frac{a}{a'} = \frac{1}{\alpha\beta} \Rightarrow m' = \left(\frac{1}{\alpha^2\beta^2}\right) m$$

So, (B) is correct Further

$$\Rightarrow a' = (\alpha\beta)a \Rightarrow \frac{v'}{t'} = (\alpha\beta) \left(\frac{v}{t}\right)$$

$$t' = \left(\frac{1}{\alpha\beta}\right) \left(\frac{v'}{v}\right) t; t' = \left(\frac{\alpha}{\beta^2}\right) t$$

So (C) is correct Further

$$\Rightarrow v' = \left(\frac{\alpha^2}{\beta}\right) v \Rightarrow \frac{\ell}{t'} = \left(\frac{\alpha^2}{\beta}\right) \left(\frac{\ell}{t}\right)$$

$$\Rightarrow \ell = \left(\frac{\alpha^2}{\beta}\right) \left(\frac{t'}{t}\right) \ell$$

$$\Rightarrow \ell' = \left(\frac{\alpha^2}{\beta}\right) \left(\frac{\alpha}{\beta^2}\right) \ell \Rightarrow \ell' = \left(\frac{\alpha^3}{\beta^3}\right) \ell$$

So (A) is also correct

Finally we have

$$p' = m'v'$$

$$p' = \left(\frac{1}{\alpha^2\beta^2}\right) m \left(\frac{\alpha^2}{\beta}\right) v$$

$$\Rightarrow p' = \left(\frac{1}{\beta^3}\right) mv = \left(\frac{1}{\beta^3}\right) p$$

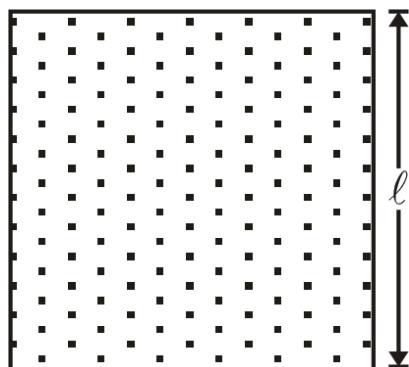
So (D) is also correct

2. Ans (A,B,C)

Given that, volume of the box = 1 m^3

Pressure of the gas = 100 Nm^{-2}

Let T be the temperature of the gas. Time between two consecutive collisions with one wall is $\Delta t = \frac{1}{500} \text{ sec.}$



This time is equal to $\frac{2l}{v_{\text{rms}}}$ where l is the side of the cube.

$$\Rightarrow \Delta t = \frac{2l}{v_{\text{rms}}} = \frac{1}{500}$$

$$\Rightarrow v_{\text{rms}} = 1000 \text{ ms}^{-1} \{ \because l = 1 \text{ m} \}$$

$$\Rightarrow \sqrt{\frac{3RT}{M}} = 1000$$

$$\Rightarrow T = \frac{(1000)^2 M}{3R} = \frac{(10)^6 (4 \times 10^{-3})}{3 \left(\frac{25}{3} \right)} = 160 \text{ K}$$

$$\text{Average kinetic energy per atom} = \frac{3}{2} k_B T$$

$$\Rightarrow \langle E \rangle = \frac{3}{2} (1.38 \times 10^{-23}) 160 \text{ J}$$

$$\langle E \rangle = 3.312 \times 10^{-21} \text{ J}$$

$$\text{Since, } PV = nRT = \left(\frac{m}{M} \right) RT$$

So, mass of helium gas in the box is given by

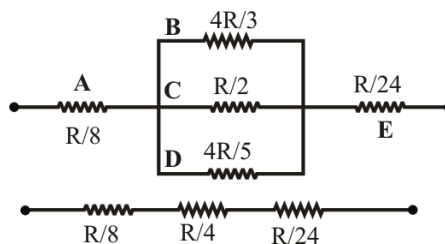
$$m = \frac{PVM}{RT}$$

Substituting the value, we get

$$m = 3 \times 10^{-4} \text{ kg}$$

3. Ans (A,B,C,D)

In steady state : (Heat In) = (Heat Out). So A is true



option (B) is also true because same drop occurs for B,C,D but R_C is minimum hence current through C is more than current through B or D.

$$(\text{Heat current}) = Q = \frac{\Delta T}{R}$$

Since, heat current (Q) is the same and R_E is minimum. So ΔT is minimum

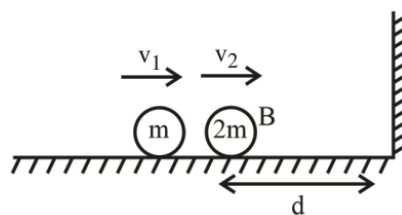
Hence, option (C) is true

$$\text{So, } Q_E = \frac{\Delta T}{\frac{4R}{3}}, Q_C = \frac{\Delta T}{\frac{4R}{2}} \text{ and } Q_D = \frac{\Delta T}{\frac{4R}{5}}$$

$$Q_B + Q_D = Q_C$$

So (D) is also true.

4. Ans (A,B,C)



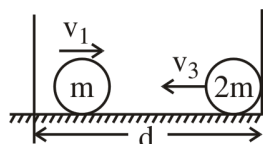
$$mv = mv_1 + 2mv_2 \quad \dots(1)$$

$$\frac{v_2 - v_1}{v} = \frac{1}{4} \quad \dots(2)$$

$$v_1 = \frac{v}{6}; v_2 = \frac{5v}{12}$$

time taken by B to collide with the wall

$$\frac{d}{v_2} = \frac{12d}{5v}$$



$$\frac{v_3}{v_2} = e \Rightarrow v_3 = \frac{5v}{48}$$

$$\text{distance between A \& B} = d - v_1 \left(\frac{12d}{5v} \right) = \frac{3d}{5}$$

$$\text{Time for collision} = \frac{d_{\text{rel}}}{V_{\text{rel}}} = \frac{3d/5}{v_1 + v_3} = \frac{3d(48)}{5 \times 13v}$$

$$\text{Distance of } 2m \text{ from wall at this instant} = V_3(t)$$

$$= \left(\frac{5v}{48} \right) \left[\frac{3d(48)}{5 \times 13v} \right] = \frac{3d}{13}$$

5. **Ans (C,D)**

$$E = E_m [2\cos^2(5\pi \times 10^{15}t) - 1] = E_m \cos(10\pi \times 10^{15}t)$$

$$2\pi f = 10\pi \times 10^{15} \Rightarrow f = 5 \times 10^{15} \text{ Hz}$$

$$E_{\text{photon}} = 6.6 \times 10^{-34} \times 5 \times 10^{15} = 33 \times 10^{-19} \text{ J} = 20.6 \text{ eV}$$

$$\text{K.E. of } e^- \text{ after ionization} = 20.6 - 13.6 = 7 \text{ eV}$$

$$\frac{\text{K.E. (ground state)}}{\text{K.E. (ionization)}} = \frac{13.6}{7} \approx 1.9$$

6. **Ans (A,B,C,D)**

$$\vec{F} = -q[\vec{v} \times \vec{B}] = q[z^2 - y^2]\hat{i}$$

$$F = 0, \text{ if } z = y$$

$$\text{for } y = 2z, F \neq 0$$

$$F \text{ is along +ve x axis if } z > y.$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**

At $t = 0$ inductor behaves as an insulator and

$$\text{hence } i = \frac{V}{R_{\text{net}}} = \frac{V}{7R/2}$$

now potential across inductor is

$$V_i = \left(\frac{2V}{7R}\right) 3R = \frac{6V}{7}$$

8. **Ans (A)**

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} + \frac{1}{R} \right) \Rightarrow f = R$$

$$(P) \frac{1}{f_{\text{eq}}} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right) + (\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right)$$

$$f_{\text{eq}} = R$$

$$(Q) \frac{1}{f_{\text{eq}}} = (\mu - 1) \left(-\frac{1}{R} - \frac{1}{R} \right) + (\mu - 1) \left(-\frac{1}{R} - \frac{1}{R} \right)$$

$$f_{\text{eq}} = \frac{-R}{2} = \frac{-f}{2}$$

$$(R) \frac{1}{f_{\text{eq}}} = (\mu - 1) \left(-\frac{1}{R} - \frac{1}{R} \right) + (\mu - 1) \left(\frac{1}{\infty} + \frac{1}{R} \right)$$

$$f_{\text{eq}} = -2R = -2f$$

$$(S) -\frac{1}{f_{\text{eq}}} = \frac{2}{f} - \frac{1}{f_m} = 2(\mu - 1) \left(\frac{1}{R} + \frac{1}{R} \right) + \left(\frac{2}{R} \right)$$

$$f_{\text{eq}} = \frac{-R}{4} = \frac{-f}{4}$$

9. **Ans (B)**

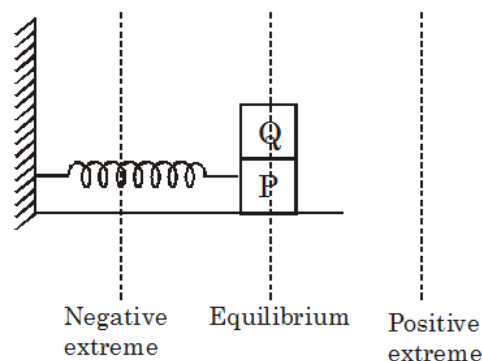
Observer in train is at rest w.r.t. train hence observer frequency and actual frequency are same (R).

For observer in-front of train all apparent frequency are more than actual frequencies (P).

For observer behind the train all apparent frequency are smaller than actual frequencies (Q).

If train approaches then all frequencies are more and when train moves away all frequencies are less (S).

10. **Ans (A)**



Friction on Q will always be towards equilibrium and friction on P will be opposite to that.

Friction is only force acting on block Q in horizontal direction which is responsible for its oscillation.

Friction force will be maximum at extreme position and zero at equilibrium.

PART-1 : PHYSICS

SECTION-II

1. **Ans (1.29 to 1.31)**

I_1 = unpolarised light intensity,

I_2 = polarised light intensity

$$I_{\text{max}} = \frac{I_1}{2} + I_2$$

$$I_{\text{min}} = \frac{I_1}{2} + 0$$

$$\frac{I_{\text{max}}}{I_{\text{min}}} = \frac{5}{3}$$

$$\left(\frac{A_{\text{max}}}{A_{\text{min}}} \right) = \sqrt{\frac{5}{3}} = 1.29$$

2. **Ans (2.50)**

$$\alpha_1 = \frac{p - p_1}{p_1 (t_1 - t)} + \alpha$$

3. Ans (0.50)

$$\left(\frac{d\theta}{dt}\right)_s = k_s(\theta - \theta_0) \quad k = \frac{4\sigma eAT_0^3}{ms}$$

$$\left(\frac{d\theta}{dt}\right)_c = k_c(\theta - \theta_0) \quad k = \frac{4\sigma eAT_0^3}{\rho sV} \alpha \frac{A}{V}$$

$$\frac{\left(\frac{d\theta}{dt}\right)_s}{\left(\frac{d\theta}{dt}\right)_c} = \frac{k_s}{k_c} = \frac{\frac{4\pi R^2}{\frac{4}{3}\pi R^3}}{\frac{6R^2}{R^3}} = \frac{3}{6} = 0.50$$

4. Ans (9.00)

$$T^2 \propto R^3 \Rightarrow T \propto R^{3/2} \Rightarrow \frac{\Delta T}{T} = \frac{3}{2} \frac{\Delta R}{R}$$

$$\text{So } \Delta T = \frac{3}{2} \left(\frac{10}{9000} \right) (90) \text{ minutes}$$

$$= \frac{3}{20} \text{ minutes} = 9 \text{ s}$$

5. Ans (1.84 to 1.86)

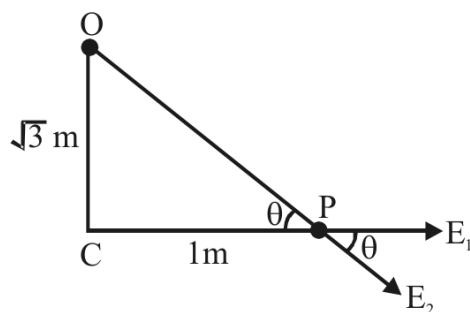
$$B = \frac{\mu_0 \epsilon_0 A}{2\pi r} \frac{dE}{dt}$$

$$B = 1.85 \times 10^{-18} \text{ T}$$

6. Ans (9.00)

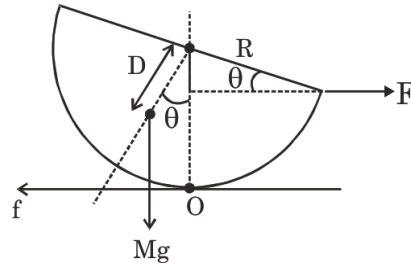
$$E_r = \vec{E}_1 + \vec{E}_2$$

$$E_r = \sqrt{E_1^2 + E_2^2 + 2E_1E_2 \cos \theta}$$



$$\text{where } E_1 = \frac{kQ}{1^2}, E_2 = \frac{kQ}{(2)^2} \text{ \& } \theta = 60^\circ$$

7. Ans (6.00)



$$D \Rightarrow \frac{4R}{3\pi}$$

$$f \Rightarrow F = \mu mg$$

Balance torque of F and Mg about 'O'.

$$F(R - R\sin\theta) = Mg D \sin\theta$$

$$\mu MgR(1 - \sin\theta) = mg \left(\frac{4R}{3\pi} \right) \sin\theta$$

$$1 - \sin\theta = \sin\theta$$

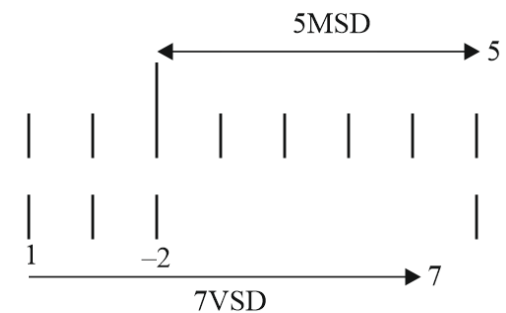
$$\sin\theta = \frac{1}{2}$$

$$\theta = 30^\circ = \frac{\pi}{6}$$

8. Ans (4.68)

$$|e| = 7VSD - 5MSD = 7(VSD - MSD) + 2MSD$$

$$= -7LC + 2MSD \Rightarrow 2MSD - 7LC$$



$$\text{Length} = 45 \text{ mm} + 5 \times 0.1 \text{ mm} + |e|$$

$$45 \text{ mm} + 5 \times 0.1 \text{ mm} + 2 \text{ mm} - 7 \times 0.1 \text{ mm}$$

$$47 - 0.2 \text{ mm} = 46.8 \text{ mm} = 4.68 \text{ cm}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (B,D)**

$$\text{Average rate of reaction} = -\frac{1}{3} \left(\frac{\Delta P_{H_2}}{\Delta t} \right)$$

$$= \frac{1}{3} \left(\frac{380/760}{100} \right) = \frac{5}{3000} \text{ atm/sec}$$

$$\text{Average rate of formation of } NH_3(g) = \frac{1}{300} \text{ atm/sec}$$

$$\frac{\text{Rate}_{N_2}}{1} = \frac{\text{Rate}_{NH_3}}{2}$$

$$\frac{\text{Rate}_{N_2}}{1} = \frac{\text{Rate}_{H_2}}{3}$$

$$\log \left(\frac{-\Delta P_{H_2}}{\Delta t} \right) = \log \left(\frac{-\Delta P_{N_2}}{\Delta t} \right) + \log 3$$

2. **Ans (B,D)**

(A) Since, the complex is $[Ti(Cp)_2(N_2)_2]$, its name would be

bis(cyclopentadienyl)bis(dinitrogen)titanium(II)

(B) Due to synergic bonding, NN bond order decreases

(C) Due to synergic bonding with Ti, NN bond order decreases further than N_2

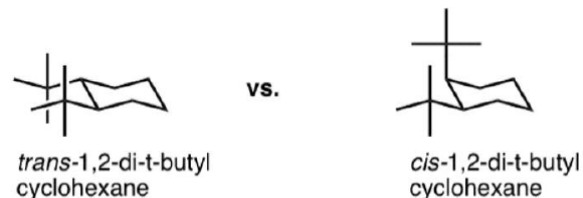
(D) $EAN = 22 - 2 + [(2 \times 6) + (2 \times 2)] = 36$.

3. **Ans (A,B,C,D)**

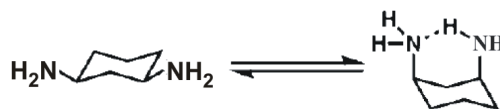
Theory based.

4. **Ans (A,B,C,D)**

(A) Trans form is less stable than the cis despite the fact that both groups are equatorial in the trans. [Significant "1,2" strain (van der Waals strain)]



(B) & (C) H bonding has stabilization effect on otherwise unstable axial conformer



(D) twist boat is more stable due to:

- less eclipsed interaction of "seat bonds"
- Less repulsion due to flagpole interaction

5. **Ans (B,C)**

5 → 6-β-D-fructofuranoside linkage

5 → 6-β-fructofuranoside linkage polymeric part

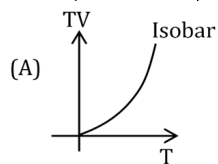
5 → 1-β-D-fructofuranosyl-α-D-glucopyranoside

or

1 → 5-α-D-glucopyranosyl-β-D-

fructofuranoside [bottom row]

6. Ans (A, B, C, D)

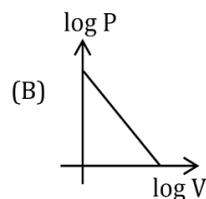


$$PV = nRT$$

$$V = \frac{nR}{P} \cdot T$$

$$V \cdot T = K \cdot T^2$$

$$y = kx^2$$



$$PV = nRT$$

$$T = \text{constant}$$

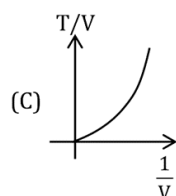
$$P = nRT \frac{1}{V}$$

$$\log P = \log(nRT) + \log \frac{1}{V}$$

$$\log P = \log k - \log V$$

$$\log P = -\log V + \log K$$

$$y = mx + C \text{ (Slope = -ve)}$$



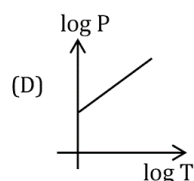
$$PV = nRT$$

$$T = \text{Constant}$$

$$P = nRT \times \frac{1}{V}$$

$$\frac{P}{V} = nRT \cdot \frac{1}{V^2}$$

$$y = kx^2$$



$$PV = nRT$$

$$V = \text{Constant}$$

$$P = \frac{nR}{V} \cdot T$$

$$\log P = \log T + \log \frac{nR}{V}$$

$$y = mx + C$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (D)

P-1, Q-2,4, R-3,5, S-3,5

8. Ans (A)

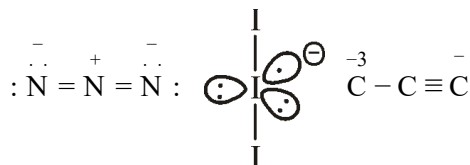
P-3,4, Q-5, R-2,5, S-1,5

9. Ans (A)

Strongly basic nucleophiles (II, III) cannot be used in presence of protic solvents (P, T)

Neutral Nucleophiles (IV) work better in presence of less protic solvents or aprotic solvents

10. Ans (B)

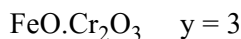
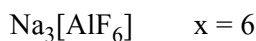


O_3^- = ozonide

PART-2 : CHEMISTRY

SECTION-II

1. Ans (754.00)

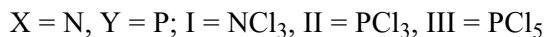


$$= 600 + 150 + 4 = 754$$

2. Ans (391.30)

$$x = 3 ; y = 8 ; z = 11 (z_1 = 4 ; z_2 = 7)$$

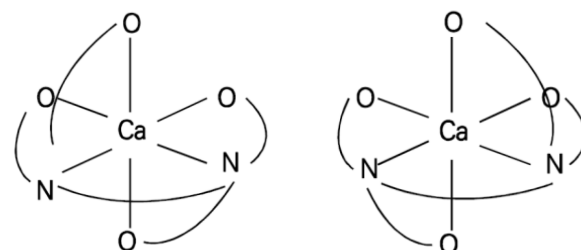
3. Ans (8.00)



The hydrolysis products are:



4. Ans (552.00)

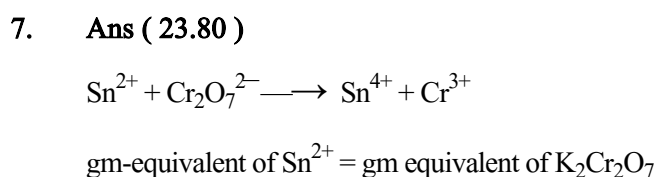
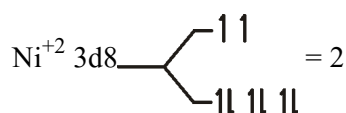


5. **Ans (548.00)**
 $500 + 30 + 18 = 548$

6. **Ans (8.00)**
 $N = \frac{2s^2}{11} \frac{2p^3}{1111} = 3$

$$S_2 = \pi \cdot 3p_y^1 = \pi \cdot 3p_z^1 = 2$$

$$O_2^- = \pi \cdot 2p_y^2 = \pi \cdot 2p_z^1 = 1$$



$$\text{Mole of } Sn^{2+} \times 2 = \frac{4.9}{294} \times \frac{20}{1000} \times 6$$

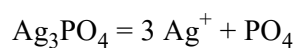
$$\text{Mole of } Sn^{2+} = 10^{-3}$$

$$\text{Mass of Sn} = 10^{-3} \times 119$$

$$\% \text{Mass of Sn} = \frac{119 \times 10^{-3}}{0.5} \times 100 = 23.8$$

8. **Ans (4.32)**
 $\lambda = K \times \frac{1000}{N}$
 $1.50 \times 10^{-4} \times 10^4 = 9 \times 10^{-6} \times 10^{-2} \times \frac{1000}{N}$
 $N = 6 \times 10^{-5}$

$$S = M = \frac{N}{n_f} = \frac{6 \times 10^{-5}}{3} = 2 \times 10^{-5} \text{ mol / L}$$



$$3S \quad S$$

$$K_{sp} = (3S)^3 \cdot S = 27 \cdot S^4 = 27 \times (2 \times 10^{-5})^4$$

$$= 4.32 \times 10^{-18}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B,C,D)**

(A) Let $P(t_1)$ and $Q(t_2)$ be two points on the parabola $y^2 = 4ax$ with focus at $S(a,0)$ Now T is the point of intersection of tangents at $P(t_1)$ and $Q(t_2)$

$$\Rightarrow T(at_1t_2, a(t_1 + t_2))$$

$$\text{Now } ST^2 = a^2(t_1 + 1)(t_2 + 1)$$

$$\text{similarly } SP \cdot SQ = a^2(t_1 + 1)(t_2 + 1)$$

(B) Let $P(x_1, y_1)$

$$\text{equation of normal } y = mx - 2am - am^3$$

$$\Rightarrow am^3 + (2a - x_1)m + y_1 = 0 \quad \begin{cases} m_1 \\ m_2 \\ m_3 \end{cases}$$

$$m_1 + m_2 + m_3 = 0$$

$$\text{Now } A(am_1^2 - 2am_1) \quad B(am_2^2 - 2am_2) \quad \& \quad C(am_3^2 - 2am_3)$$

$$\text{equation of circle } x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{so } a^2m^4 + (4a^2 + 2ag)m^2 - 4afm + c = 0$$

$$\Rightarrow m_1 + m_2 + m_3 + m_4 = 0$$

$$\Rightarrow m_4 = 0 \quad (\because m_1 + m_2 + m_3 = 0)$$

$$\Rightarrow c = 0$$

(C) Let the three points on the parabola

$$\text{be } P(at_1^2, 2at_1), Q(at_2^2, 2at_2), R(at_3^2, 2at_3)$$

Point of intersection of tangents drawn at these points are $A(at_2t_3, a(t_2 + t_3))$, $B(at_1t_3, a(t_1 + t_3))$, $C(at_1t_2, a(t_1 + t_2))$

for finding orthocentre of triangle ABC

$$m_{BC} = \frac{a(t_2 - t_3)}{at_1(t_2 - t_3)} = \frac{1}{t_1}$$

slope of altitude through vertex A is $-t_1$

equation of altitude through A is

$$y - a(t_2 + t_3) = -t_1x + at_1t_2t_3 \dots (1)$$

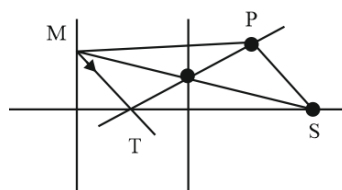
similarly equation of altitude through B is

$$y - a(t_1 + t_3) = -t_2x + at_1t_2t_3 \dots (2)$$

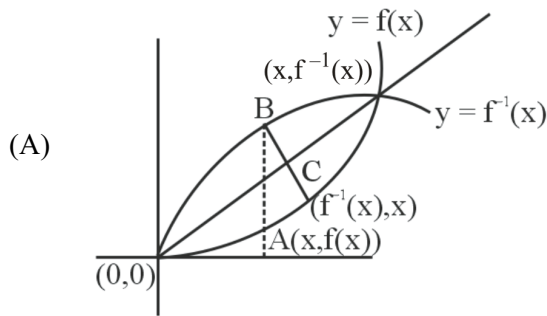
subtracting (2) from (1), we get

$$a(t_1 - t_2) = x(t_2 - t_1) \Rightarrow x = -a$$

(D)



2. Ans (A,B,C,D)



slope of OC > slope of OA

$$\frac{x}{f^{-1}(x)} > \frac{f(x)}{x} \Rightarrow x^2 > f^{-1}(x) f(x) \cdot \forall x \in (0, 1)$$

(B) Here $f(\alpha) = f(\beta)$

$\Rightarrow f'(x) = 0$ has at least one real $\gamma \in (\alpha, \beta)$

$f(x)$. $f'(x) = 0$ has roots α, γ, β

and its derivative $(f'(x))^2 + f(x)f''(x) = 0$ has at least two real root in (α, β) .

(C) Here $g(a) = g(b) = 0$

let $x = c$ is a point of global maxima $C \in (a, b)$

then $g'(c) = 0$ $g''(c) < 0$

but $g''(c) + g'(c) \cdot g(c) = g(c)$

$g''(c) = g(c) < 0$

since $g(a) = g(b) = 0$

$\Rightarrow$ maximum value of $g(x) = 0 \quad \forall x \in (a, b)$

(D) Here consider $g(x) = f^2(x) f(1-3x) (1-2x)$

apply R.T. on $\left[\frac{1}{3}, \frac{1}{2}\right]$

3. Ans (A,B)

$$(2\lambda - 2)2 - 2(1 - 2\lambda) - 1(3 - \lambda) = 0$$

$$\Rightarrow \lambda = 1 \Rightarrow F(1, 1, -1)$$

The normal to the plane π_1 and π_2

$$= \vec{n}_1 \times \vec{n}_2 = i + 15j + 16k$$

Equation of planes are

$$1(x - 3) + 15(y + 1) + 16(z + 2) = 0$$

$$1(x + 1) + 15(y - 3) + 16(z - 0) = 0$$

$$\Rightarrow x + 15y + 16z + 44 = 0, x + 15y + 16z = 44$$

4. Ans (A,B,C)

E and H are confocal foci $(\pm 1, 0)$

focus of parabola is $(2, 3)$

$\therefore$ Equation of circle is

$$((x + 1)(x - 1) + y^2) + \lambda y = 0$$

$$\therefore x^2 + (y - 2)^2 = 5$$

5. Ans (A,C)

$$I_{n+k} = \int_{\frac{n+k}{2}}^{\frac{n+k+1}{2}} \frac{\sin(\pi \sin^2 \pi x)}{(\sqrt{2})^x} dx \Rightarrow x = \frac{k}{2} + t$$

$$I_{n+k} = \int_{n/2}^{\frac{n+1}{2}} \frac{\sin\left(\pi \sin^2 \pi \left(\frac{k}{2} + t\right)\right)}{(\sqrt{2})^{\frac{k}{2} + t}} dt$$

$$I_{n+k} = \frac{1}{2^{k/4}} \int_{n/2}^{\frac{n+1}{2}} \frac{\sin(\pi \sin^2 \pi t)}{(\sqrt{2})^t} dt \Rightarrow \frac{I_{n+k}}{I_n} = 2^{-k/4}$$

6. Ans (A,B,C)

$$f(x) = \begin{cases} 0, & \text{if } x < -2 \\ 1 - |x + 1|, & \text{if } -2 \leq x \leq 0 \\ 1 - |x - 1|, & \text{if } 0 \leq x \leq 2 \\ 0, & \text{if } x > 2 \end{cases}$$

5 points.

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

B. $2x + y - 1 = 0$ is axis and $(1, -1)$ is the point of intersection with the tangent at vertex, length of latus rectum is $\sqrt{5}$.

$$h = 1 \pm \left(\frac{\sqrt{5}}{4}\right) \left(-\frac{1}{\sqrt{5}}\right) = \frac{3}{4}, \frac{5}{4}$$

$$k = -1 \pm \left(\frac{\sqrt{5}}{4}\right) \left(\frac{2}{\sqrt{5}}\right) = \frac{-1}{2}, -\frac{3}{2}$$

C. $f(x) > f(2) \quad \forall x < t$

$$x \rightarrow 2^- \Rightarrow f(x) f(2) = 1$$

$$a^2 - 9a - 10 \geq 0 \Rightarrow a \leq -1 \text{ or } a \geq 10$$

$$a \in [10, \infty)$$

If $a=10$, $x=2$ is critical point

$$D. (\cos x - 2)(\cos x - (a - 3)) \geq 0$$

$$\Rightarrow \cos x \leq a - 3 \quad \therefore a \geq 4$$

8. Ans (D)

$$A. f(9) = f(4+5) = f(4 \times 5) = f(20) = f(16+4)$$

$$= f(16 \times 4) = f(64) = f(8 \times 8) = f(16) = f(4 \times 4)$$

$$= f(4+4) = f(8) = 9$$

$$B. xy \leq \left(\frac{x+y}{2}\right)^2 = 1 \Rightarrow 0 < xy \leq 1$$

$$x^3 y^3 (x^3 + y^3) = 2(xy)^3 ((x+y)^2 - 3xy)$$

$$= 2(xy)^3 (4 - 3xy) \Rightarrow \text{put } t = xy$$

$$3^3 (4 - 3t) \leq \left(\frac{4 - 3t + 3t}{4}\right)^4 = 1$$

$$\therefore x^3 y^3 (x^3 + y^3) \leq 2$$

$$C. [\{x\}] = 0 \Rightarrow f(x) = (3 - x^7)^{1/7}$$

$$f(f(x)) = x$$

$$D. \left(\frac{1 - 3\tan^2 1^\circ}{3 \tan 1^\circ - \tan^3 1^\circ}\right) \frac{1}{\cos 3^\circ} = \frac{1}{\tan 3^\circ \cdot \cot 3^\circ} = 1$$

9. Ans (C)

$$B. \int_1^e \frac{x^4 \ln x + x^2 - x^2 + 2}{x^3 \ln x + x} dx$$

$$= \int_1^e x dx - \int_1^e \frac{\frac{1}{x} - \frac{2}{x^3}}{\ln x + \frac{1}{x^2}} dx$$

$$C. 2 \tan^{-1} (\sec^2 \pi x) \geq \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} (x^3 - x^2 + x + 2) \leq \frac{\pi}{2}$$

$$\therefore x^3 - x^2 + x + 1 = 0$$

No integral solution

$$D. f(x) > x^2 \forall x \in \mathbb{R} - \{\pm 1\}$$

$$f(x) > f^2(x) > x^4 \forall x \in \mathbb{R} - \{\pm 1\}$$

$$f(f(x)) > f^2(x) > x^4 \forall x \in \mathbb{R} - \{\pm 1, 0\}$$

$$f(f(f(x))) \leq x^8 \text{ has solution only at } x = \pm 1$$

10. Ans (A)

$$a. D.R \text{ of line } -5\hat{i} + \hat{j} + 3\hat{k}$$

$$\frac{x-3}{-5} = \frac{y-2}{1} = \frac{z-1}{3}$$

$$N(-5\lambda + 3, \lambda + 2, 3\lambda + 1)$$

$$MN \perp (-5\hat{i} + \hat{j} + 3\hat{k})$$

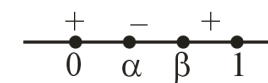
$$\Rightarrow -5(-5\lambda + 2) + (\lambda - 7) + 3(3\lambda - 6) = 0$$

$$\Rightarrow \lambda = 1 \text{ so point}$$

$$N(-2, 3, 4)$$

b. f has period 1 and is discontinuous at

$$x = 0, 1, 2, \dots, 10.$$



$$f'(x) = 2\pi \cos 2\pi x + 1$$

f has local maxima at $x = \alpha$

$$c. \int_0^{\pi/2} x \cdot \frac{\sin x}{x} dx = (-\cos x)_0^{\pi/4} = \frac{2 - \sqrt{2}}{2}$$

$$p + q = 4$$

$$d. \int_0^e x \cdot x^{\frac{1}{2}} \cdot x^{\frac{1}{2} \cdot \frac{1}{3}} \cdot x^{\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4}} \dots dx$$

$$= \int_0^e x \left(1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots \infty - 1\right)$$

$$\int_0^e x^{e-1} dx = e^{e-1} \Rightarrow A + B = 0$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (177.00)

$$\frac{720!}{6!} = \frac{720(719)!}{720} = (719)!$$

Number of zero's at the end of (719)! = exponent

of 5 in (719)!

$$\left[\frac{719}{5}\right] + \left[\frac{719}{25}\right] + \left[\frac{719}{125}\right] + \left[\frac{719}{625}\right]$$

$$= 143 + 28 + 5 + 1 = 177$$

2. Ans (100.00)

$$\log^{\sin x + \cos x} = -1 \Rightarrow \sin x \cdot \cos x = 2^{-1} \Rightarrow \sin 2x = 1$$

$$\log_2^{\sin x + \cos x} = \frac{1}{2}(\log_{10}^n - 1)$$

$$\log_2^{\sin x + \cos x} = \log_{10}^n - 1$$

$$\log_2^{1 + \sin 2x} = \log_{10}^n - 1 \Rightarrow \log_{10} n = 2 \quad \therefore n = 100$$

3. Ans (667.00)

$${}^{2000}C_{1998} b^{1998} a^2 = {}^{2000}C_{1997} b^{1997} a^3$$

$$\Rightarrow b = 666a$$

$$\therefore a = 1, b = 666 \text{ and } a+b=667$$

4. Ans (200.00)

$$2000x^6 + 100x^5 + 10x^3 + x - 2$$

$$= (20x^2 + x - 2)(100x^4 + 10x^2 + 1)$$

$$20x^2 + x - 2 = 0 \Rightarrow x = \frac{-1 \pm \sqrt{161}}{40}$$

$$\therefore m + n + r = -1 + 161 + 40 = 200$$

5. Ans (6.00)

Divide P(x) by Q(x) the remainder is $x^2 - x + 1$

$$P(z_i) = z_i^2 - z_i + 1$$

$$\sum_{i=1}^4 P(z_i) = \sum_{i=1}^4 (z_i^2 - z_i + 1) = 6$$

6. Ans (251.00)

$$2 \cos a \sin b = \sin(a+b) - \sin(a-b)$$

$$\sum_{K=1}^n 2 \cos(K^2 a) \sin(Ka)$$

$$= \sum_{K=1}^n (\sin(K(K+1)a) - \sin(K(K-1)a))$$

$$= \sin(n(n+1)a)$$

$$\therefore \sin\left(\frac{n(n+1)\pi}{2008}\right) \text{ is an integer.}$$

This can be only $\{-1, 0, 1\}$

$$\therefore \frac{2n(n+1)}{2008} \in \mathbb{I}$$

$$1004 = 2^2 \times 251 \text{ divides } n(n+1)$$

$$\Rightarrow 251 \text{ divides } n, n+1$$

$$\therefore n = 251$$

7. Ans (5.00)

$$C_{28} = \sum_{k=1}^5 a_{2k} b_{k8} = \sum_{k=1}^5 \frac{2}{3k-2} \times \frac{8}{3k+1}$$

$$= \frac{16}{3} \left(1 - \frac{1}{16}\right) = 5$$

8. Ans (0.33 or 0.34)

$$\frac{dy}{dx} - \frac{1}{\sqrt{x^2-1}} y = x$$

$$\text{Integrating factor} = \frac{1}{x + \sqrt{x^2-1}}$$

$$\therefore y \cdot \frac{1}{x + \sqrt{x^2-1}} = \int \frac{x dx}{x + \sqrt{x^2-1}} + c$$

$$\frac{y}{x + \sqrt{x^2-1}} = \frac{x^3}{3} - \frac{1}{3} (x^2-1)^{\frac{3}{2}} + c$$

$$\text{Give } y(1) = 1 \Rightarrow C = \frac{2}{3}$$

$$\therefore y(-1) = -\frac{1}{3} \Rightarrow |y(-1)| = \frac{1}{3}$$

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	D	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B	B,C	B,C	B,C	B,C,D	A,B,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	1	9	9	3	5	2	3	9

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	D	C	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,D	A,C	C,D	A	A	A		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	1	6	4	7	5	3	8	4

PART-3 : MATHEMATICS

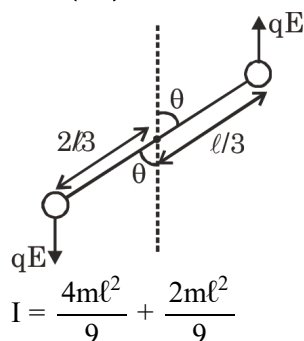
SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	A	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C,D	A,B,C	A,C,D	B,D	A,B,C	A,B		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	8	5	2	6	4	3	2	3

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (B)**

Apply Bernaulli's theorem between just below points of tubes

$$P_0 + \rho gh_1 + \frac{1}{2}\rho v^2 = P_0 + \rho gh_2 + 0$$

 2. **Ans (A)**


$$I = \frac{6ml^2}{9}$$

$$\tau_{\text{net}} = qEl \sin \theta$$

$$\tau_{\text{net}} = I\alpha$$

 for small oscillation $\alpha = \omega^2 \theta$

$$qEl\theta = \frac{2ml^2}{3}\alpha \Rightarrow \frac{3qE\theta}{2ml} = \omega_{\text{phase}}^2$$


 F_{net} toward centre

$$T - qE = m\omega_{\text{particle}}^2 R \quad [\omega_{\text{particle}} = \theta_{\text{max}} \omega_{\text{phase}}]$$

3. **Ans (D)**

$$\frac{1}{R_{eq}} = \frac{1}{2R} + \frac{1}{R}$$

$$R_{eq} = \frac{2}{3}R$$

$$C_{eq} = \frac{C}{2} + C = \frac{3}{2}C$$

$$\tau = R_{eq} C_{eq} = RC$$

$$i = C \frac{dV}{dt}$$

$$\frac{i_1}{i_2} = \frac{C_1}{C_2} = \frac{1}{2} \quad \dots(1)$$

$$i_1 + i_2 = \frac{3\varepsilon}{2R} e^{-\frac{t}{\tau}} \quad \dots(2)$$

Solving (1) & (2)

4. **Ans (A)**

$$i = JA$$

$$J = ne V_d$$

$$i = Ane V_d$$

$$i = \text{constant}$$

$$AV_d = \text{Constant}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,B)**

Common potential

$$V_C = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} = 14 \text{ V}$$

$$q_A = C_1 V_C = 28 \mu\text{C}$$

$$\Delta q = C_1 V_1 - C_1 V_C = 12 \mu\text{C}$$

Heat produced

$$H = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2 = 60 \mu\text{J}$$

6. **Ans (B,C)**

Since current is leading voltage so, it is RC circuit

$$\tan \phi = \frac{1}{R\omega C}; \quad \text{where } \phi = 60^\circ$$

$$\text{Solving, } C = 10 \mu\text{F}$$

7. **Ans (B,C)**

Flux linked with the loop

$$\phi = \vec{B} \cdot \vec{S}$$

induced emf,

$$e = \frac{d\phi}{dt}$$

8. **Ans (B,C)**

$$L.C. = \frac{1\text{mm}}{100} = 0.01 \text{ mm}$$

$$+ve \text{ zero error} = 0.05 \text{ mm}$$

$$\text{diameter of wire} = 2 \text{ mm} + 0.45 \text{ mm} - 0.05 \text{ mm}$$

$$= 2.40 \text{ mm}$$

$$\text{surface area} = 2\pi r l = \frac{22}{7} \times \frac{2.40}{10} \text{ cm} \times 4.2 \text{ cm}$$

$$= 3.168 \text{ cm}^2$$

in appropriate significant figure

$$\text{surface area} = 3.2 \text{ cm}^2$$

9. **Ans (B,C,D)**

$$\lambda_{K\alpha} > \lambda_{K\beta} = \frac{1}{\lambda_{K\alpha}} < \frac{1}{\lambda_{K\beta}}$$

$$\text{So, } \frac{1}{\lambda_{K\alpha}} = 3.3(\text{nm})^{-1} \Rightarrow \lambda_{K\alpha} = 0.303 \text{ nm}$$

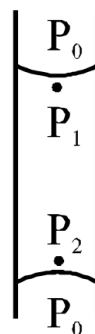
$$\frac{1}{\lambda_{K\alpha}} = R(z-1)^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$$

$$\Rightarrow 3.3 \times 10^9 \text{ m}^{-1} = 1.1 \times 10^7 \times (z-1)^2 \times \frac{3}{4}$$

$$\Rightarrow z-1 = 20 \Rightarrow z = 21$$

$$V_a = \frac{1240}{\lambda_c (\text{nm})} \cdot V = 12400 \text{ V}$$

10. **Ans (A,B,C)**



$$P_1 = P_0 - \frac{2T_s}{r} = (P_0 - 600) \frac{\text{N}}{\text{m}^2}$$

$$P_2 = P_1 + \rho gh = P_0 - 600 + 10^4 h$$

$$\text{for } h = 4 \text{ cm, } P_2 = P_0 - 200 \frac{\text{N}}{\text{m}^2}$$

$$P_2 < P_0$$

$$\text{for } h = 6 \text{ cm, } P_2 = P_0$$

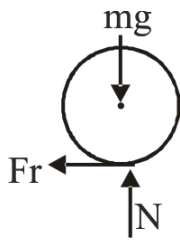
$$\text{for } h = 8 \text{ cm, } P_2 = P_0 + 200 \frac{\text{N}}{\text{m}^2}$$

$$P_2 > P_0$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (1)**



$$F_r = ma_{cm}$$

$$R F_r = I a$$

$$I = \frac{1}{2} m R^2$$

$$N = mg$$

$$F_r = \mu N$$

$$a_{cm} = \mu g$$

$$\alpha = 2\mu g / R$$

$$v_0 - \mu g t = \left[\frac{2\mu g}{R} t - \omega_0 \right] R$$

$$t = 1 \text{ sec.}$$

2. **Ans (9)**

$$E_{n \rightarrow m} : E_{m \rightarrow 2} : E_{2 \rightarrow 1} = 5 : 72 : 243$$

$E_{m \rightarrow 2} \Rightarrow$ Transition corresponding to visible region.

$$E_{n \rightarrow 2} : E_{2 \rightarrow 1} = 77 : 243$$

$$\Rightarrow \left(\frac{1}{2^2} - \frac{1}{n^2} \right) : \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = 77 : 243 \Rightarrow n = 9.$$

3. **Ans (9)**

Mass defect in the reaction

$$\Delta m = 2 \times 2.015 - 3.07 - 1.009 = 0.004 \text{ amu}$$

$$\text{Fractional mass defect} = \frac{0.004}{2 \times 2.015}$$

mass defect in 1 kg

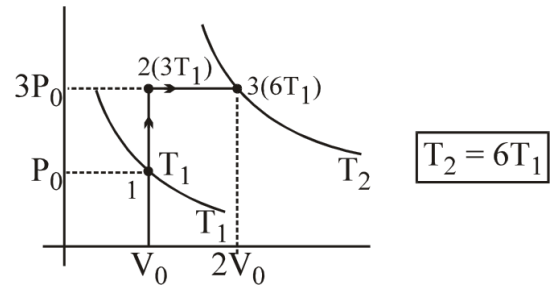
$$m = \frac{0.004}{4.03} \times 1 \text{ kg}$$

Energy released

$$E = mc^2 = 8.9 \times 10^{13} \text{ J}$$

4. **Ans (3)**

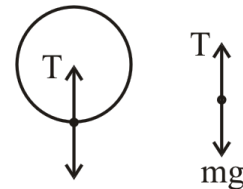
$$\Delta Q = nC_V 2T_1 + nC_P (3T_1)$$



$$\Delta T = 5T_1$$

$$\therefore C = \frac{\Delta Q}{n\Delta T} = \frac{2C_V + 3C_P}{5} = \frac{2 \times (5/2)R + 3 \times (7/2)R}{5} = 3.1 R$$

5. **Ans (5)**



From FBD, we have

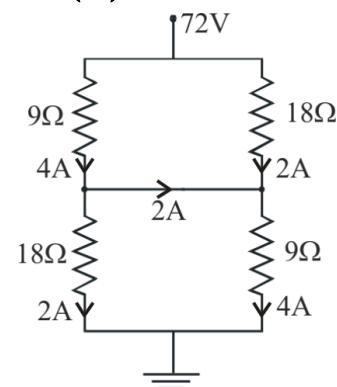
$$T - mg = \frac{mv^2}{r}$$

$$52 - 2 = \frac{0.2}{0.1} v^2$$

$$v^2 = 25$$

$$v = 5 \text{ m/s}$$

6. **Ans (2)**



7. **Ans (3)**

$$v = \sqrt{\frac{\lambda x^2 L (3g)}{L(2)\lambda x}}$$

$$v = \sqrt{\frac{3gx}{2}}$$

$$a = \frac{3g}{4}$$

8. Ans (9)

$$F = -\frac{\partial U}{\partial r} = -\frac{\partial}{\partial r} \left(-\frac{ke^2}{3r^3} \right) \&$$

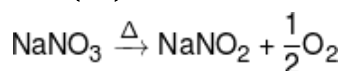
$$\frac{ke^2}{r^4} = \frac{mv^2}{r^4} \dots\dots(i)$$

$$mvr = \frac{nh}{2\pi} \dots\dots(ii)$$

PART-2 : CHEMISTRY

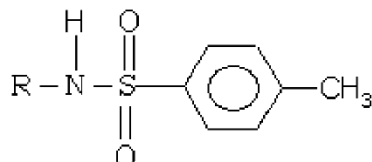
SECTION-I (i)

1. Ans (B)

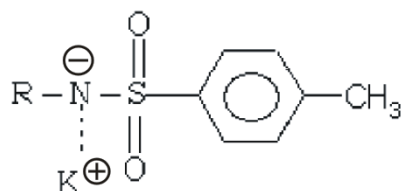


2. Ans (D)

After formation of N substituted sulphonamide



with the Hinsbergs reagent, then solubility of the salt



depends on the length of hydrophobic Rportion.

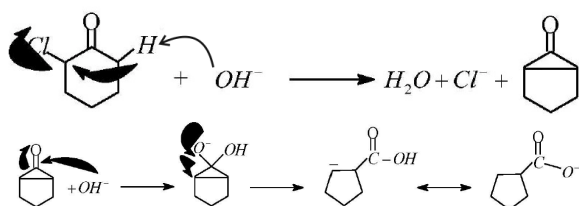
A very long chain may cause

non- dissolutions and the amine may therefore appear as secondary

3. Ans (C)

When α- haloketones are treated with alkali, ring contraction occurs (Favorsky rearrangement)

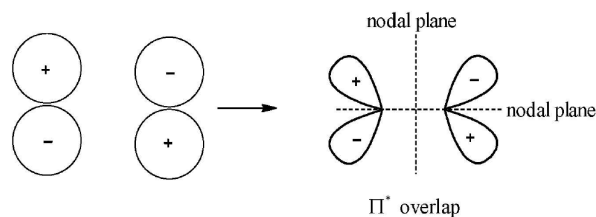
Mech:



PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (B,D)



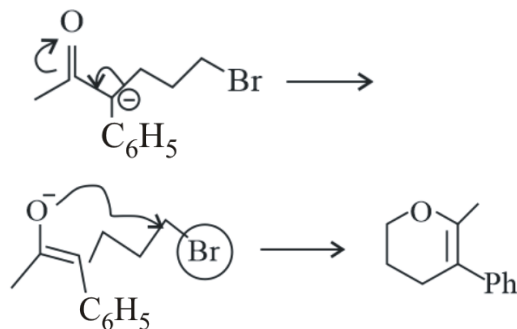
6. Ans (A,C)

It is dsp^2 hybridized due to pairing up because of strong field ligand CO. It cannot show ionization isomerism

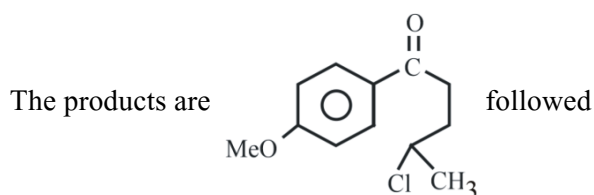
7. Ans (C,D)

Solvated electrons gets associated

8. Ans (A)



9. Ans (A)



by reduction and EAS

10. Ans (A)

$M^+ + I^- \rightarrow MI$, $E^0 = 0.952.V$ (from the given data)

$$\Rightarrow 0.952. = -0.059 \log K_{sp}$$

PART-2 : CHEMISTRY

SECTION-III

2. Ans (6)

$$\log k_{\text{uncatalysed}} = \log A - \frac{E_a}{500RT}$$

$$\log k_{\text{cat}} = \log A - \frac{E_{a_{uc}}}{400RT}$$

$$\frac{E_{a_{uc}}}{5} = \frac{E_a}{4}$$

$$4E_{a_{uc}} = 5E_a$$

$$E_{a_{uc}} - E_a = 30$$

$$\frac{5}{4}E_a - E_a = 30$$

$$E_a = 120\text{kJ}$$

$$E_{a_{uc}} = 150\text{kJ}$$

$$150 \div 25 = 6$$

3. Ans (4)

$$10^2 \times 2 \times 10^{-2} = n \times 25/3 \times 300$$

$$n = 0.8$$

$$\text{Let Neon} = x \text{ gm}$$

$$\text{Argon} = y \text{ gm}$$

$$\frac{x}{20} + \frac{y}{40} = 0.8$$

$$2x + y = 32$$

$$\text{Given } x + y = 28\text{gm}$$

$$2x + 28 - x = 32$$

$$x = 4\text{gm}$$

4. Ans (7)

Heat released to cool 500 g water from 20°C to 0°C

$$Q = ms\Delta T$$

$$= 500 \times 4.214 \times 20 = 42140\text{J} = 42.14\text{kJ}$$

Number of moles of water (Ice) that will melt to absorb

$$\frac{42.14}{6.02} = 7$$

Number of cubes of ice that will melt = 7

5. Ans (5)

I → application of Huckel's rule of periphery electrons

Total no. of periphery electrons participating in aromatic character is 18

II → non-aromatic

III → $14\pi e^-$ - aromatic

IV → $14\pi e^-$ - aromatic

V → non-aromatic

VI → Anti-aromatic

VII → non-aromatic

VIII → Anti-aromatic

IX → aromatic $10\pi e^-$

X → aromatic $6\pi e^-$

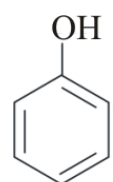
6. Ans (3)

In Hoffmann bromide reaction, cross over products are not obtained.

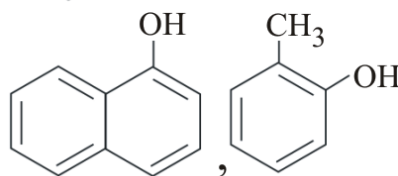
7. Ans (8)

CO_2 , SiO_2 , N_2O_3 , NO_2 , P_4O_6 , P_4O_8 , P_4O_9 , P_4O_{10}

8. Ans (4)



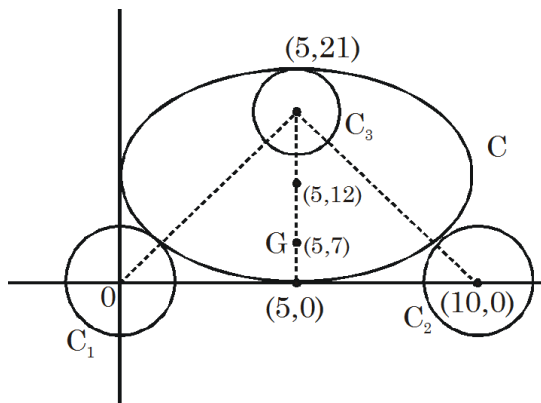
$\text{CH}_3\text{CO}_2\text{Na}$,



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)



Equation of C is

$$(x-5)^2 + (y-12)^2 = 12^2 \quad \dots(1)$$

this also touch at (5,0) centroid lie on the circle

$$(x-5)^2 + (y-12)^2 = 5^2$$

2. Ans (A)

Observe

$$A(x) = 24({}^x C_1)$$

$$L(x) = 24({}^{x+1} C_2)$$

$$E(x) = 24({}^{x+2} C_3)$$

$$N(x) = 24({}^{x+3} C_4)$$

$$\text{So, } {}^1 C_1 + {}^2 C_1 + \dots + {}^{10} C_1 = {}^{11} C_2$$

$${}^2 C_2 + {}^3 C_2 + \dots + {}^{11} C_2 = {}^{12} C_3$$

$${}^3 C_3 + {}^4 C_3 + \dots + {}^{12} C_3 = {}^{13} C_4$$

$$\text{and } {}^4 C_4 + \dots + {}^{13} C_4 = {}^{14} C_5$$

3. Ans (A)

$$\int_{-1}^4 f(x) dx = 10$$

$$\Rightarrow \int_{-1}^0 f(x) dx + \int_0^4 f(x) dx = 10$$

$$\Rightarrow -\int_0^1 f(x) dx + \int_0^4 f(x) dx = 10$$

$$\Rightarrow \int_0^4 f(x) dx = \frac{23}{2}$$

Therefore Required Area = 23

4. Ans (B)

Number of paths passing through M are

$$= \frac{7}{2} \times \frac{4}{2} = 126$$

Number of paths through N are

$$= \frac{6}{2} \times \frac{5}{3} = 150$$

$$150 + 126 - 90 = 186$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (B,C,D)

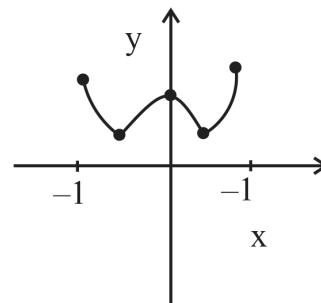
$$\text{Put } z = e^{i\theta}$$

$$|1-z| = 2 \left| \sin \frac{\theta}{2} \right| \text{ and } |1+z^2| = \sqrt{(1+\cos 2\theta)^2 + \sin^2 2\theta}$$

$$= 2 \left| 1 - 2\sin^2 \frac{\theta}{2} \right|$$

$$\text{Hence, } y = |x| + |1-2x^2| \in \left[\frac{\sqrt{2}}{2}, 2 \right]; x \in [-1, 1]$$

$$\text{where } x = \sin \frac{\theta}{2}$$



$$\text{Hence, } |1-z| + |1+z^2| \in [\sqrt{2}, 4]$$

6. Ans (A,B,C)

$$\alpha^2 - \alpha - 1 = 0$$

$$\Rightarrow \alpha^n - \alpha^{n-1} - \alpha^{n-2} = 0$$

$$\text{Similarly } \beta^n - \beta^{n-1} - \beta^{n-2} = 0$$

$$\text{Hence, } T(n) = T(n-1) + T(n-2)$$

So, $T(n)$ forms fibonacci sequence

$$\{1, 1, 2, 3, 5, \dots\}.$$

$$\sum_{n=1}^{2021} T(n) = \frac{1}{\sqrt{5}} ((\alpha + \alpha^2 + \dots + \alpha^{2021}) - (\beta + \dots + \beta^{2021}))$$

$$= \left(\frac{\alpha^{2023} - \beta^{2023}}{\sqrt{5}} \right) - \left(\frac{\alpha - \beta}{\sqrt{5}} \right)$$

$$= T(2023) - 1$$

$$T(10) = 55$$

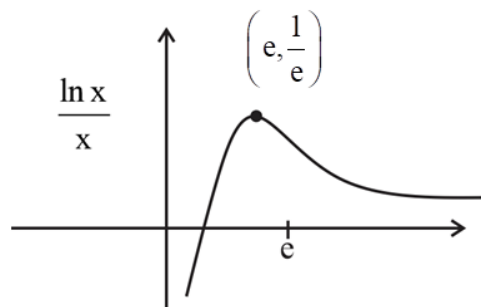
7. **Ans (A,C,D)**

$$\lim_{x \rightarrow 0} \frac{2022 \left((1+x^{2022})^{\frac{1}{2021}} - 1 \right)}{x} - \frac{2023 \left((1+x^{2023})^{\frac{1}{2021}} - 1 \right)}{x} = \frac{1}{2021}$$

Hence, $P = 2021$

$$\text{Let } f(x) = \frac{\ln x}{x}$$

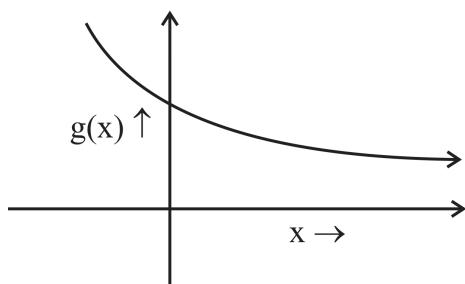
Hence, $f(x) = 2021$ has no solution



$$\text{If } g(x) = \left(\frac{2018}{2021} \right)^x + \left(\frac{2019}{2021} \right)^y + \left(\frac{2020}{2021} \right)^z$$

then $g(x)$ is strictly decreasing function

Hence, $g(x) = 1$ has exactly one real solution.



8. **Ans (B,D)**

$$f(x) + f(1-x) \geq 2\sqrt{f(x)f(1-x)}$$

by A.M - G.M inequality

$$\begin{aligned} \int_0^1 f(x) dx &= \int_0^{\frac{1}{2}} f(x) dx + \underbrace{\int_{\frac{1}{2}}^1 f(x) dx}_{\text{Put } x=1-t} \\ &= \int_0^{\frac{1}{2}} f(x) dx + \int_0^{\frac{1}{2}} f(1-x) dx \\ &= \int_0^{\frac{1}{2}} (f(x) + f(1-x)) dx \geq \int_0^{\frac{1}{2}} 2 dx \geq 1. \end{aligned}$$

9. **Ans (A,B,C)**

$x = -1$ is the only value in domain of f

$$f(-1) = \frac{\pi}{2} \Leftrightarrow g\left(\frac{\pi}{2}\right) = -1$$

Now, verify the options

10. **Ans (A,B)**

Observe that

$$A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow A^K \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and $\text{Trace}(AB) = \text{Trace}(BA)$. Hence D is wrong

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (8)**

$$S = 1 \times 1 + \left(2 \times \frac{1}{2} \right) +$$

$$\dots\dots + 2^{2023} \left(\frac{1}{2^{2023}} \right) = 2024$$

2. **Ans (5)**

$$\int_0^1 f(x) dx = \int_0^1 2t f(t^2) dt \text{ by substitution}$$

$$x = t^2$$

The given equation

$$\int_0^1 2t f(t^2) dt = \int_0^1 (f(t^2))^2 dt + \int_0^1 t^2 dt$$

$$\int_0^1 (f(t^2) - t)^2 dt = 0, \text{ Since } f \text{ is continuing}$$

$$\text{So, } f(t^2) = t \quad \forall t \in [0, 1]$$

$$\text{Put } t = \frac{1}{2} \Rightarrow \boxed{f\left(\frac{1}{4}\right) = \frac{1}{2}}$$

3. Ans (2)

$$\int_0^1 e^{x+\sqrt{x}} dx = 2 \int_0^1 x e^{x^2+x} dx$$

by letting $x = y^2$

$$\text{So, } I = \int_0^1 (1+2x) e^{x^2+x} dx = e^{x^2+x} \Big|_0^1 = e^2 - 1$$

4. Ans (6)

$$\frac{x^2}{16} + \frac{y^2}{1} = 1$$

$$a^2 = 16, b^2 = 1$$

$$a = 4, b = 1$$

let $A(x_1, y_1)$ and $B(x_2, y_2)$ then P, Q will be

$$P \left(x_1, \frac{y_1}{4} \right) \text{ and } Q \left(x_2, \frac{y_2}{4} \right)$$

$$\text{mid point of AB will be } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

mid point of PQ will be

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{8} \right) = (h, k)$$

$$\Rightarrow \text{mid point of AB will be } (h, 4k)$$

let point $S(t, t+8)$ be a point on $y = x + 8$

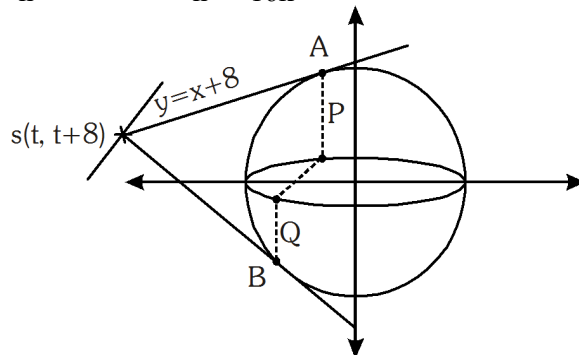
AB is chord of contact of $x^2 + y^2 = 16$

$$\Rightarrow xt + y(t+8) = 16 \quad \dots(i)$$

AB as chord whose mid point is $(h, 4k)$

$$xh + y(4k) = h^2 + 16k^2 \quad \dots(ii)$$

$$\frac{t}{h} = \frac{t+8}{4k} = \frac{16}{h^2 + 16k^2}$$



$$\Rightarrow t = \frac{16h}{h^2 + 16k^2} \text{ and } t+8 = \frac{64k}{h^2 + 16k^2}$$

$$\Rightarrow \frac{16h}{h^2 + 16k^2} + 8 = \frac{64k}{h^2 + 16k^2}$$

$$\Rightarrow 2h + (h^2 + 16k^2) = 8k$$

$$\Rightarrow x^2 + 16y^2 + 2x - 8y = 0$$

$$x^2 + \alpha y^2 + \beta x + \gamma y = 0$$

$$\Rightarrow \alpha = 16, \beta = 2, \gamma = -8$$

$$\Rightarrow \alpha - \beta + \gamma = 16$$

5. Ans (4)

Notice that α, β lies on the boundary of S, so each

lies on one of the four planes. However we can

ignore the plane defined by $(0,0,0), (2,4,0),$

$(5,1,0)$ (as it lowest).

Let P_1 be planes by $(2,4,0), (5,1,0), (3,2,10)$

$$\text{So, } P_1 : \frac{x}{6} + \frac{y}{6} + \frac{z}{60} = 1$$

Similarly P_2 be plane by points $(0,0,0), (2,4,0),$

$(3,2,10)$

$$\text{So, } P_2 : x - \frac{y}{7} - \frac{z}{5} = 0$$

P_3 be plane by $(0,0,0), (5,1,0)$ and $(3,2,10)$

$$\text{So, } P_3 : x - 5y - \frac{7}{10}z = 0$$

Using this we find pts as $\left(\frac{12}{5}, \frac{23}{10}, 13 \right)$

and $\left(\frac{12}{5}, \frac{23}{10}, \frac{25}{4} \right)$. Only lowest is on tetrahedron

$$\text{so, } \alpha = \frac{25}{4}.$$

$$\text{Similarly } \beta = \frac{5}{2}$$

$$\text{So, } \alpha - \beta = \frac{15}{4}$$

6. Ans (3)

Let $P = xyz$

$$3x - 4y + pz = 12$$

$$-12x + py + 4z = 3$$

$$px + 12y - 3z = 4$$

$$\boxed{x = \frac{1}{3}, y = \frac{1}{4}, z = 1}$$

7. **Ans (2)**

Let angle between $\hat{u}$ and $\hat{v}$ be θ

$$\text{then } \frac{2 + 2 \cos \theta}{4} + \sin^2 \theta = 1$$

$$\Rightarrow \cos^2 \frac{\theta}{2} = \cos^2 \theta$$

$$\Rightarrow \theta = \frac{2\pi}{3}$$

$$\text{So } |\hat{u} \times \hat{v}| = \left| \frac{\hat{u} - \hat{v}}{2} \right|$$

8. **Ans (3)**

$$P(E_1) = 1 - \frac{1}{2^{n-1}}$$

$$P(E_2) = \frac{n+1}{2^n}$$

$$P(E_1 \cap E_2) = \frac{n}{2^n}$$

For being independent we must have

$$P(E_1 \cap E_2) = P(E_1) \times P(E_2)$$

$$\text{which gives } n = 3$$